# Lecture 11: Node Voltage Analysis

## EE201 – Circuit Theory I

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current voltage, node voltage $e$ supernode

# Review: two ways to solve an LTI resistive circuit

## Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

## Write its equations as it is

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages $\mathbf{e}$ as the unknowns, KVL holds automatically.

Only KCL is left to write.

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- Its steps are the same for every LTI resistive circuit.

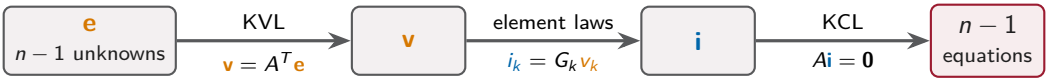
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- These simplifications do not work for every circuit.
- Node voltage analysis writes the equations of the circuit directly, without changing the circuit.
- Its unknowns are only the $n-1$ node voltages, instead of $2b$ branch voltages and currents.
- Its steps are the same for every LTI resistive circuit.
- Later, mesh current analysis will do the same with mesh currents.

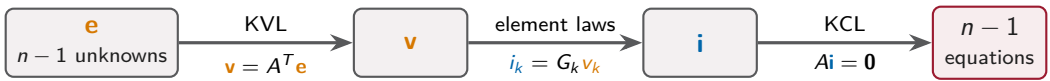
# Roadmap

1. **The idea** — node voltages as the unknowns
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection
3. **A voltage source to the ground** — a node voltage is known
4. **A voltage source between two nodes** — Method I: a new unknown current
5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

# The idea: node voltages as the unknowns

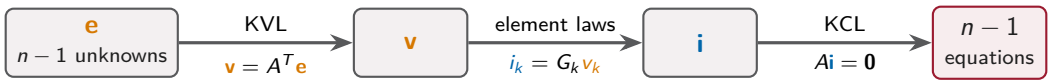


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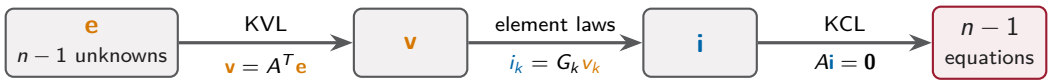

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- This gives $n-1$ equations in $n-1$ unknowns.
- Once $\mathbf{e}$ is known, $\mathbf{v} = A^T\mathbf{e}$ gives the branch voltages, and the element laws give the branch currents.

# The procedure

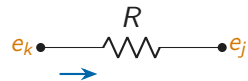
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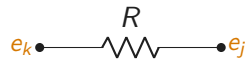
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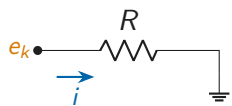

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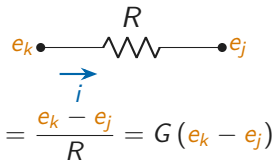

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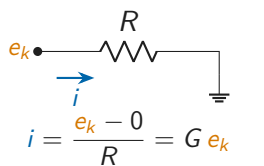



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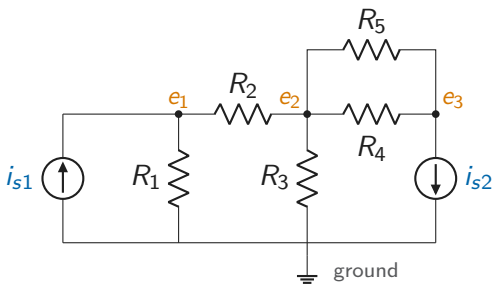




The current leaving node $k$ through a resistor is its own node voltage minus the other one, over $R$.

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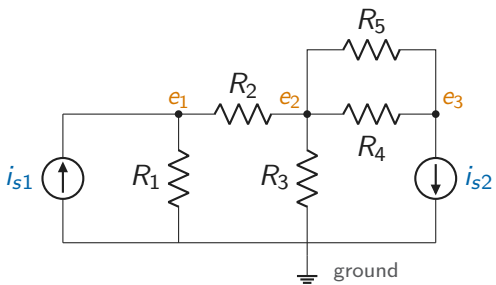
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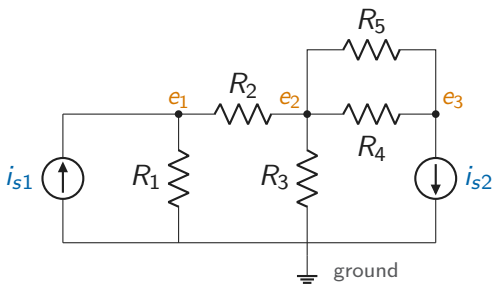



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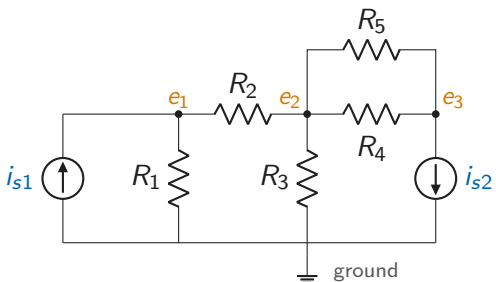

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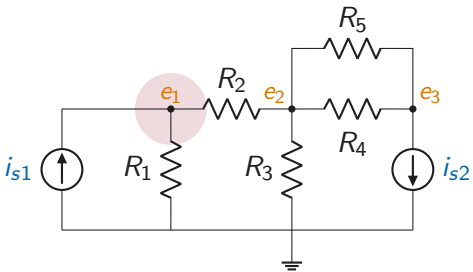
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So we need three KCL equations, at nodes 1, 2 and 3.

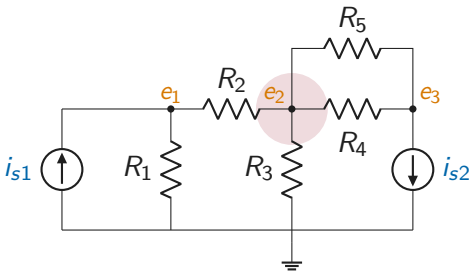
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**Node 1:**

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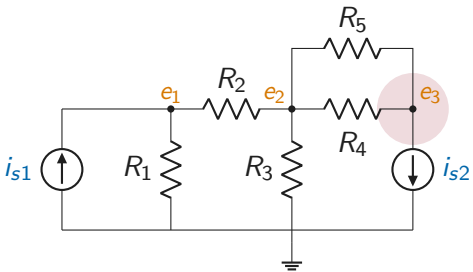

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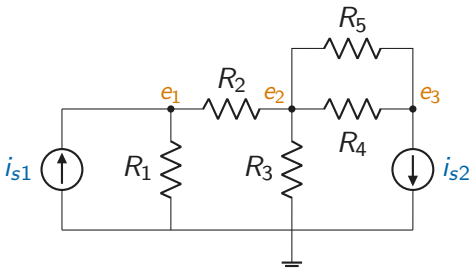
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Every term is a current leaving the node.

A source pointing into the node enters with a minus sign.

# Example 11.1: matrix form

With the conductances $G_k = 1/R_k$:

$$\begin{aligned}
(G_1 + G_2)\, e_1 - G_2\, e_2 &= i_{s1} \\
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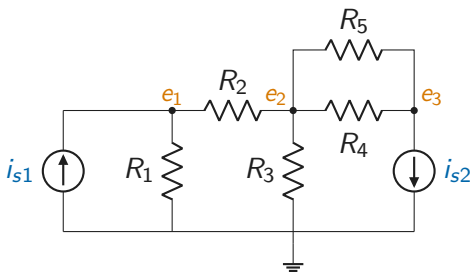
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$Y_n$ is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_s$.

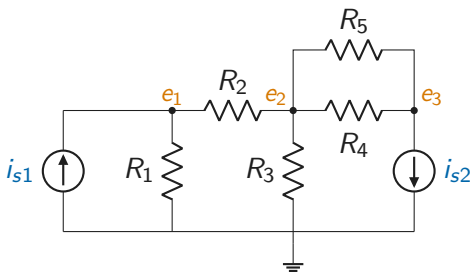
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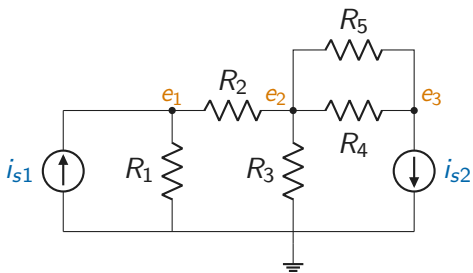



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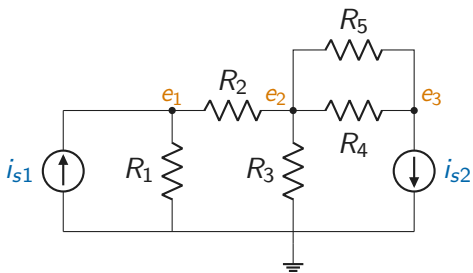

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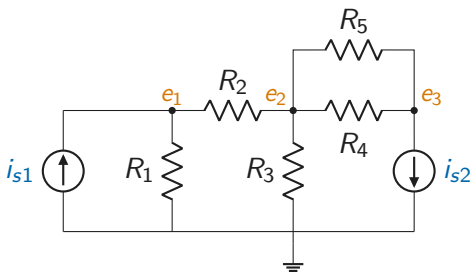
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So $Y_n$ is symmetric: the entry $(k,j)$ equals the entry $(j,k)$.

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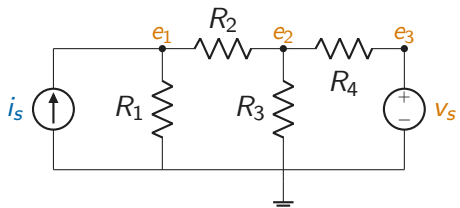
$G$ is diagonal, so $Y_n$ is symmetric.

$-A_S \mathbf{i}_S$ adds up the source currents entering each node.

# Roadmap

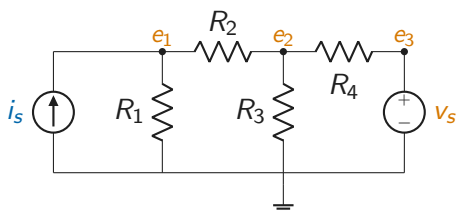
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5. **The supernode** — Method II: KCL around both nodes
6. **Dependent sources and the ground** — constraint equations, another ground

# Example 11.2: a voltage source to the ground




Unknowns $e_1$, $e_2$, $e_3$: three equations needed.

# Example 11.2: a voltage source to the ground

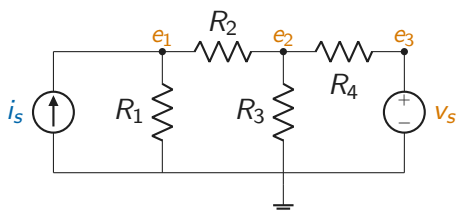



Unknowns $e_1$, $e_2$, $e_3$: three equations needed.

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

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# Example 11.2: a voltage source to the ground



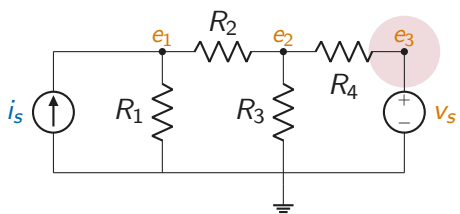

Unknowns $e_1$, $e_2$, $e_3$: three equations needed.

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

**Node 2:** $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

**Question.** What does KCL at node 3 give?

# Example 11.2: a voltage source to the ground




Unknowns $e_1$, $e_2$, $e_3$: three equations needed.

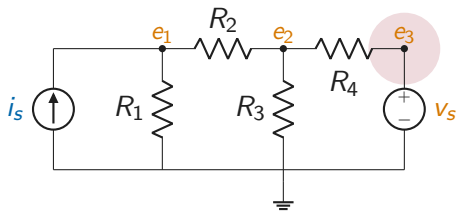
**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

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**Question.** What does KCL at node 3 give?

The current through $v_s$ is unknown: KCL gives no useful equation in $e_1$, $e_2$, $e_3$.

# Example 11.2: a voltage source to the ground




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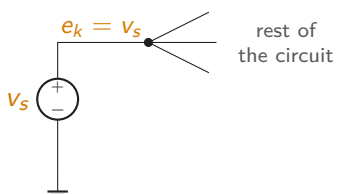
But $e_3 = v_s$: this is the third equation.

# A voltage source to the ground

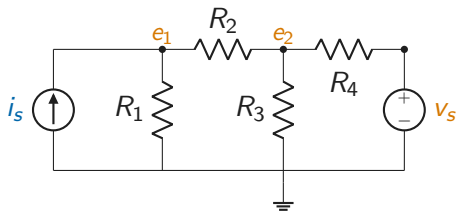
## Remark

If an independent voltage source $v_s$ is connected between the ground and node $k$, then $e_k = v_s$ is known.

So $e_k$ is not an unknown, and we do not write KCL at node $k$.



# Example 11.2, revisited

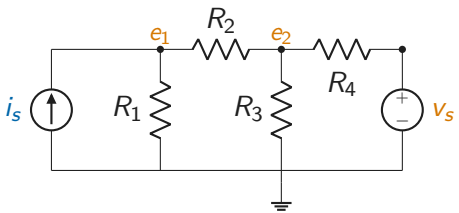



Unknowns $e_1$, $e_2$ only: two equations needed.

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

**Node 2:** $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

# Example 11.2, revisited



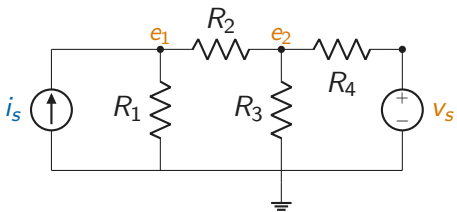

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**Node 2:** $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

# Example 11.2, revisited




Unknowns $e_1$, $e_2$ only: two equations needed.

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

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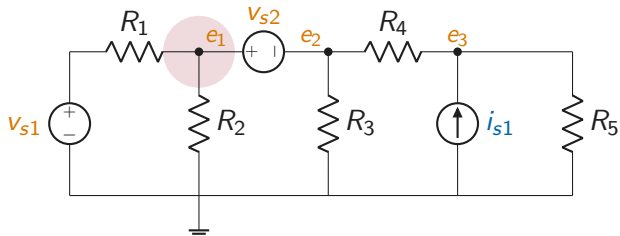
The known $v_s$ moves to the right side.

$v_s/R_4$ acts like a current source into node 2: a source transformation.

# Roadmap

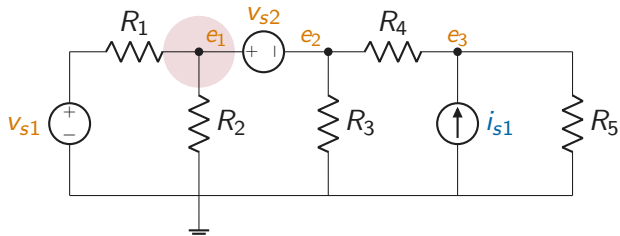
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known ✓
4. **A voltage source between two nodes** — Method I: a new unknown current
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# Example 11.3: a voltage source between two nodes



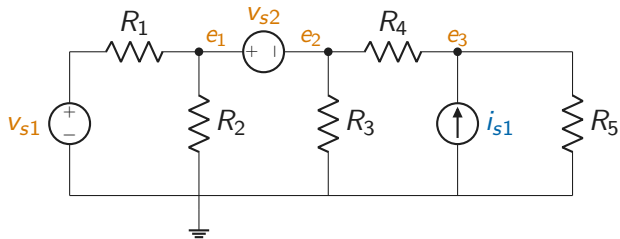

- **Question.** Write KCL at node 1: what is the current through $v_{s2}$?

# Example 11.3: a voltage source between two nodes



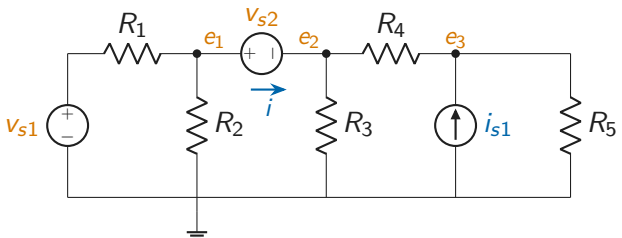

- **Question.** Write KCL at node 1: what is the current through $v_{s2}$?
- It cannot be written with $e_1$ and $e_2$: the law of a voltage source fixes its voltage, not its current.

# Example 11.3: a voltage source between two nodes




- **Question.** Write KCL at node 1: what is the current through $v_{s2}$?
- It cannot be written with $e_1$ and $e_2$: the law of a voltage source fixes its voltage, not its current.
- **Modified node voltage analysis** offers two ways out: Method I (its current is a new unknown) and Method II (a supernode).

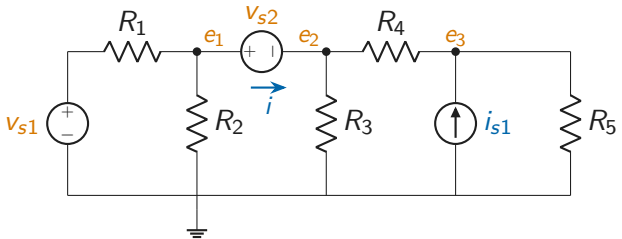
# Method I: the current as an unknown




**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Unknowns: $e_1$, $e_2$, $e_3$ and the current $i$.

# Method I: the current as an unknown

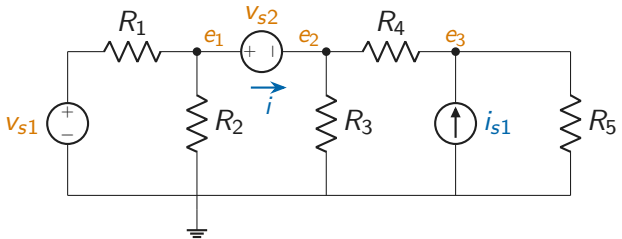



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**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Unknowns: $e_1$, $e_2$, $e_3$ and the current $i$.

# Method I: the current as an unknown



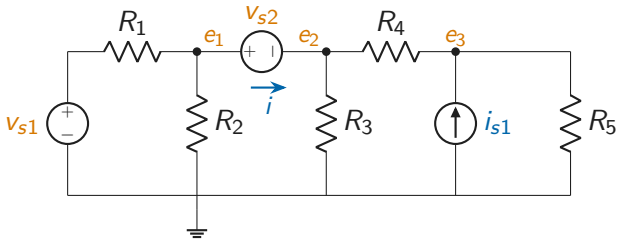

**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: $e_1$, $e_2$, $e_3$ and the current $i$.

# Method I: the current as an unknown




**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

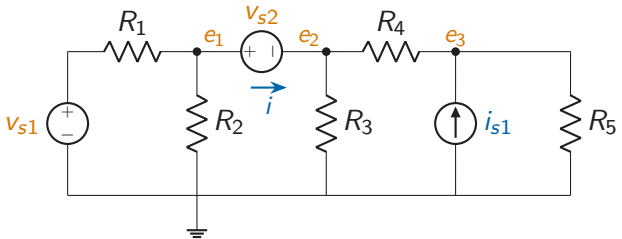
**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: $e_1$, $e_2$, $e_3$ and the current $i$.

Four unknowns, three equations: one more is needed.

# Method I: the current as an unknown




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**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Unknowns: $e_1$, $e_2$, $e_3$ and the current $i$.

Four unknowns, three equations: one more is needed.

**Constraint:** $e_1 - e_2 = v_{s2}$, the law of the source.

# Method I: matrix form

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

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- The first three rows are KCL; the last row is the constraint.

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- The first three rows are KCL; the last row is the constraint.
- The unknown current $i$ is part of the solution: we get it directly.

# Method I: matrix form

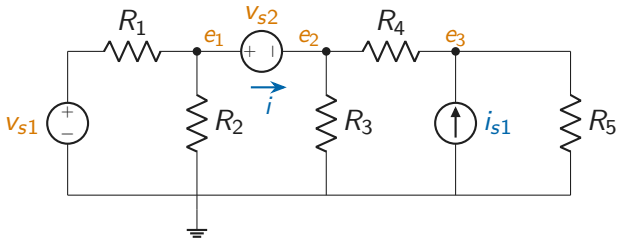
$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}, \qquad G_k = \frac{1}{R_k}$$

- The first three rows are KCL; the last row is the constraint.
- The unknown current $i$ is part of the solution: we get it directly.
- The price: one more unknown and one more equation.

# Roadmap

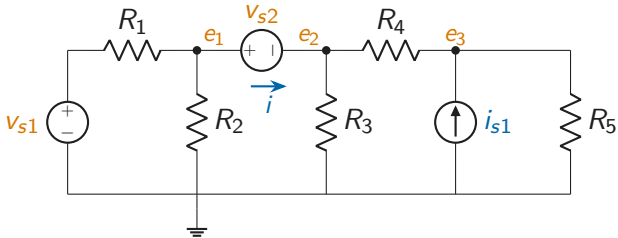
1. **The idea** — node voltages as the unknowns ✓
2. **Only current sources** — $Y_n \mathbf{e} = \mathbf{i}_s$, written by inspection ✓
3. **A voltage source to the ground** — a node voltage is known ✓
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5. **The supernode** — Method II: KCL around both nodes
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# Method II: the supernode




**Question.** Can we get rid of the unknown $i$?

# Method II: the supernode

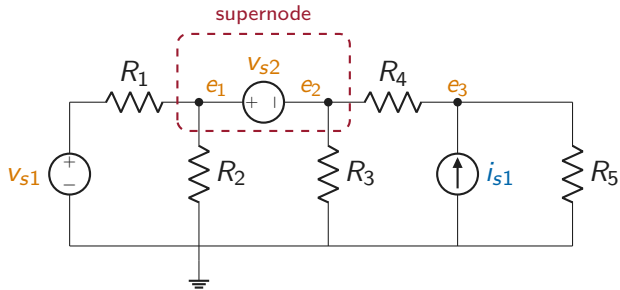



**Question.** Can we get rid of the unknown $i$?

Add the KCL equations of nodes 1 and 2.

$i$ leaves node 1 and enters node 2, so it cancels.

# Method II: the supernode



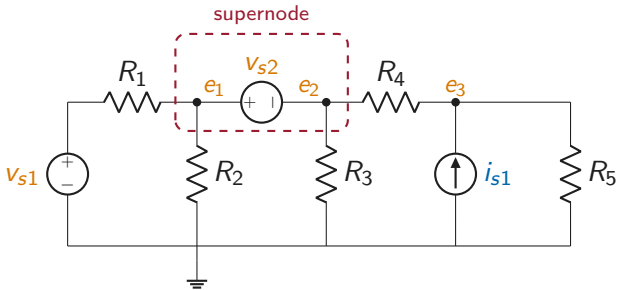

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This is KCL for a closed surface around both nodes and the source.

# Method II: the supernode




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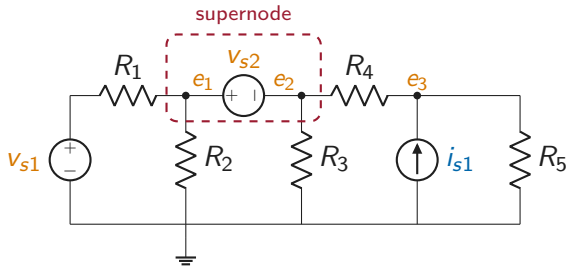
$i$ leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

## Supernode

Write KCL as if the two nodes of the voltage source were a single node.

# Method II: the equations



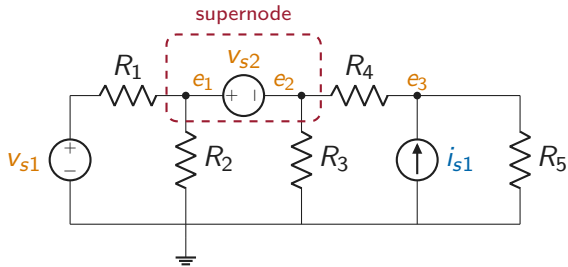

**Supernode:**

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

**Constraint:** $e_1 - e_2 = v_{s2}$

# Method II: the equations




**Supernode:**

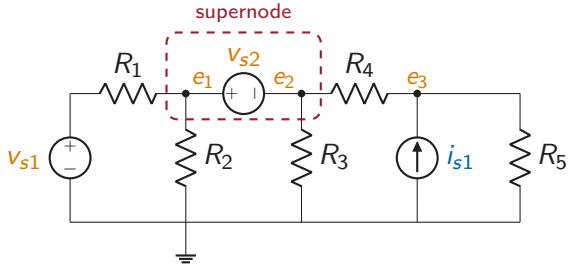
$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

**Constraint:** $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

# Method II: the equations




**Supernode:**

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

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Three unknowns, three equations.

The matrix is not symmetric any more.

# Method I or Method II?

| | **Method I** | **Method II** |
|---|---|---|
| extra unknown | the current $i$ of the source | none |
| KCL | at every non-ground node | at the supernode, and the others |
| extra equation | the constraint $e_1 - e_2 = v_{s2}$ | the same constraint |
| size (this example) | $4 \times 4$ | $3 \times 3$ |
| the current $i$ | part of the solution | from KCL afterwards |

# Method I or Method II?

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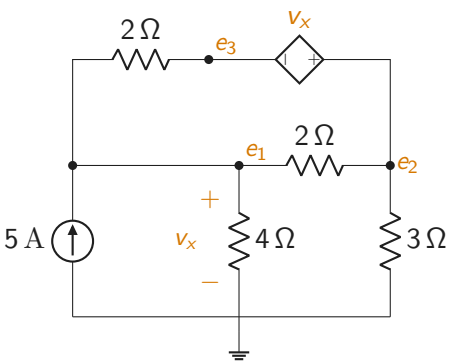
Both give the same node voltages.

The constraint is always the law of the voltage source.

# Roadmap

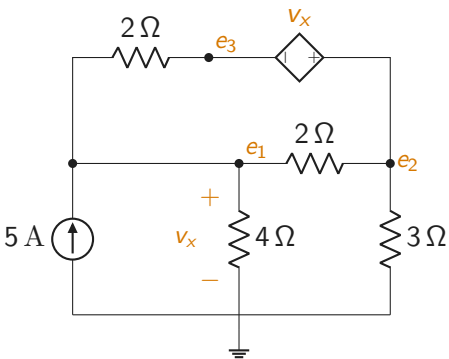
1. **The idea** — node voltages as the unknowns ✓
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5. **The supernode** — Method II: KCL around both nodes ✓
6. **Dependent sources and the ground** — constraint equations, another ground

# Example 11.4: a dependent source




**Question.** Find the power absorbed by the 3 Ω resistor.

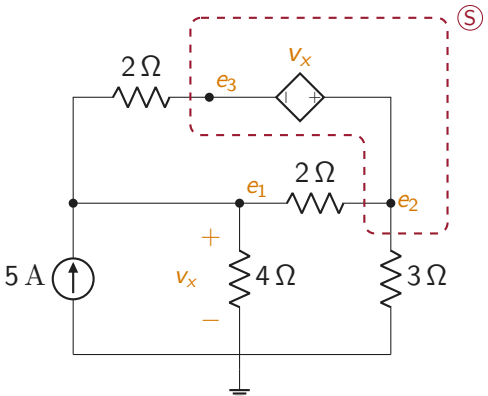
# Example 11.4: a dependent source




**Question.** Find the power absorbed by the 3 Ω resistor.

The voltage source $v_x$ is between nodes 3 and 2; neither is the ground.

# Example 11.4: a dependent source

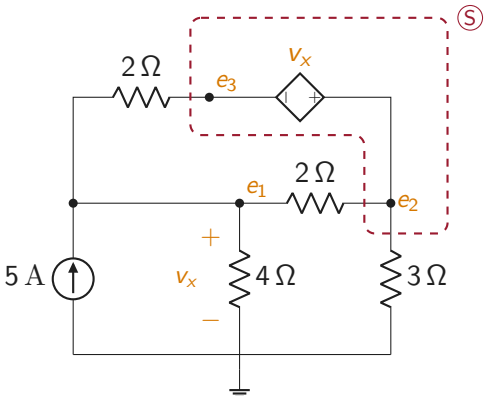



**Question.** Find the power absorbed by the 3 Ω resistor.

The voltage source $v_x$ is between nodes 3 and 2; neither is the ground.

So use a supernode (S) around nodes 2 and 3.

# Example 11.4: a dependent source




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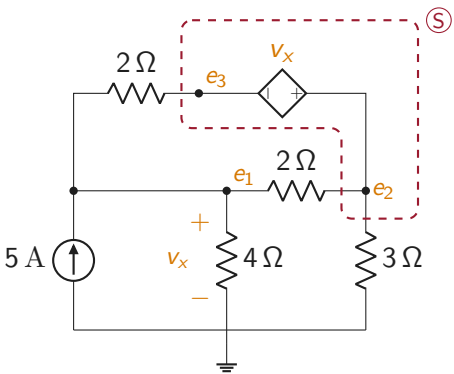
The voltage source $v_x$ is between nodes 3 and 2; neither is the ground.

So use a supernode Ⓢ around nodes 2 and 3.

Unknowns: $e_1$, $e_2$, $e_3$.

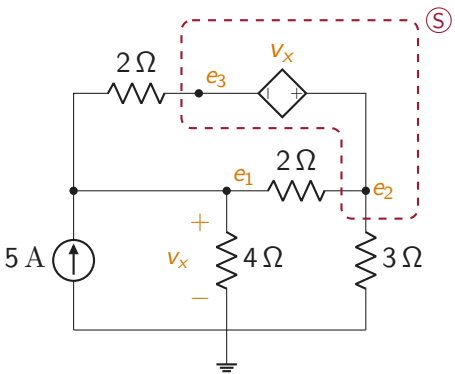
Equations: KCL at node 1, KCL at Ⓢ, and a constraint.

# Example 11.4: the equations




**Node 1:** $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

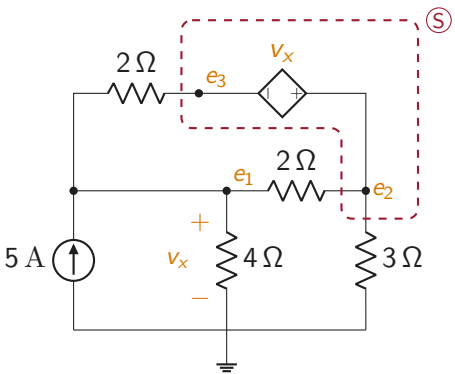
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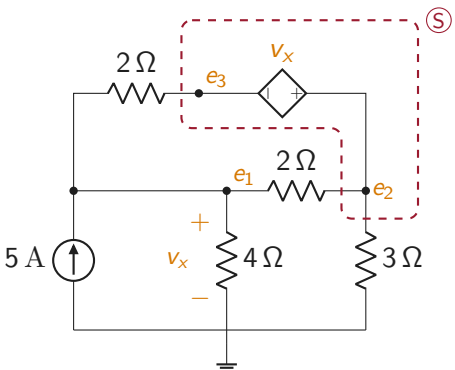



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**Question.** What is the constraint?

# Example 11.4: the equations



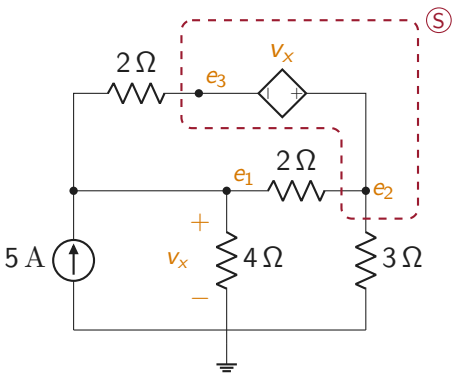

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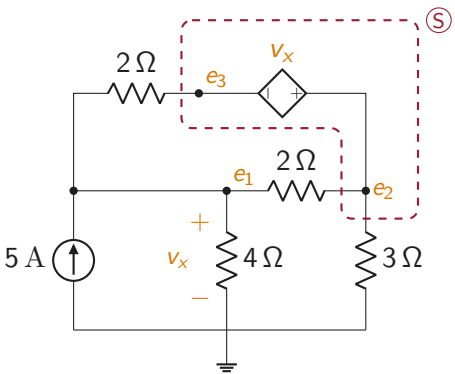
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The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

**Constraint:** $e_2 - e_3 = e_1$ (3)

Three equations, three unknowns.

# Example 11.4: solve

- (1): $\frac{5}{4}e_1 - \frac{1}{2}e_2 - \frac{1}{2}e_3 = 5$; put $e_1 = e_2 - e_3$ from (3): $\frac{3}{4}e_2 - \frac{7}{4}e_3 = 5$ (4)

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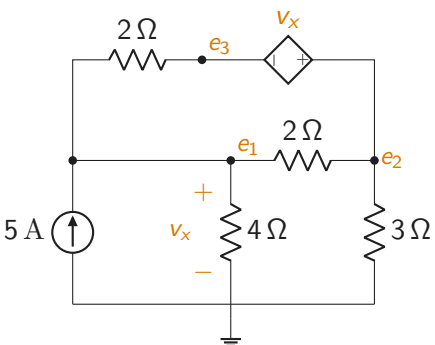
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**Remark**

If a dependent voltage source is connected between node $k$ and the ground, do not write KCL at node $k$: its current is not known.

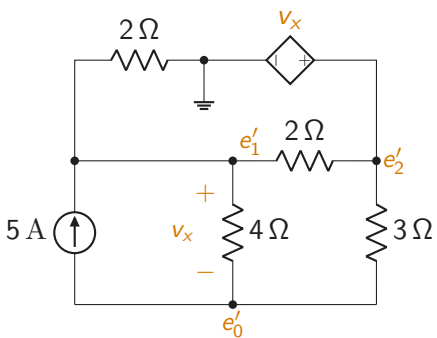
Write the constraint that the dependent source gives instead.

# Exercise: choose node 3 as the ground




**Question.** Redo the example with node 3 as the ground. Does the answer change?

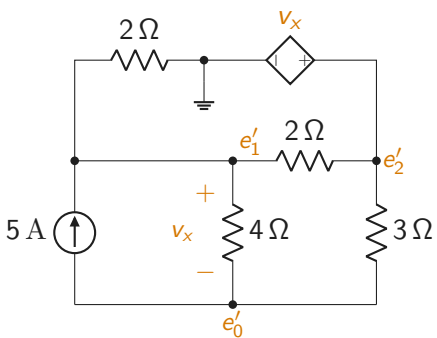
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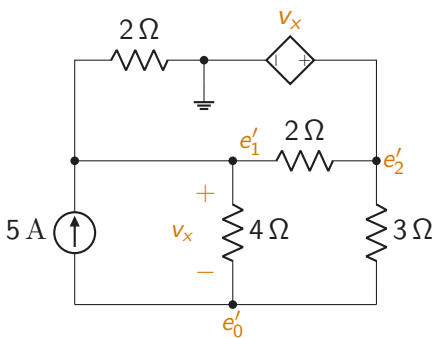



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Unknowns: $e_1'$, $e_2'$ and $e_0'$, the old ground.

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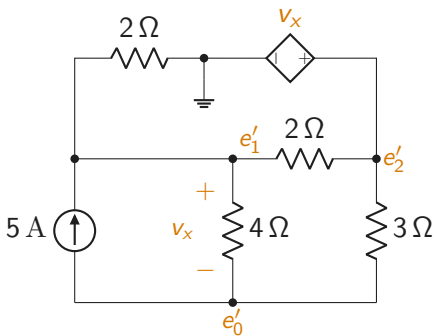
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**Constraint:** $e_2' = v_x = e_1' - e_0'$; no KCL at node 2.

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- Put the constraint into both KCL equations:

$$3e_1' + e_0' = 20, \qquad -7e_1' + 11e_0' = -60.$$

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Tip: put the ground at one end of a voltage source; this saves a supernode.

# Summary

| In the circuit | What to do |
|---|---|
| resistors and current sources | KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_s$, by inspection; $Y_n$ is symmetric |
| a voltage source from the ground to node $k$ | $e_k = v_s$ is known; no KCL at node $k$ |
| a voltage source between two nodes | Method I: its current as an unknown, plus the constraint<br>Method II: KCL at the supernode, plus the constraint |
| a dependent source | write its control variable with node voltages: a constraint |
| the choice of ground | any node; the branch voltages and currents do not change |

After the node voltages: $\mathbf{v} = A^T\mathbf{e}$, then the currents from the element laws.